Transferring secure-messaging's SCKA correctness bound to the extracted SPQR code

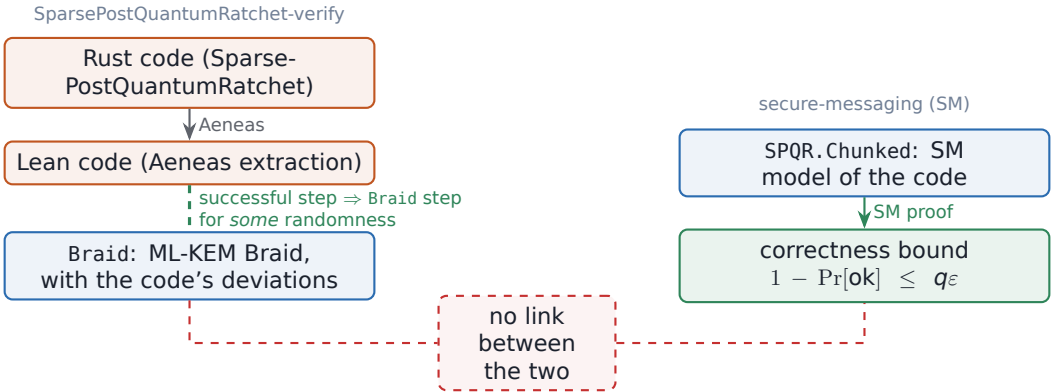
SpqrBridge slides, the method

SpqrBridge

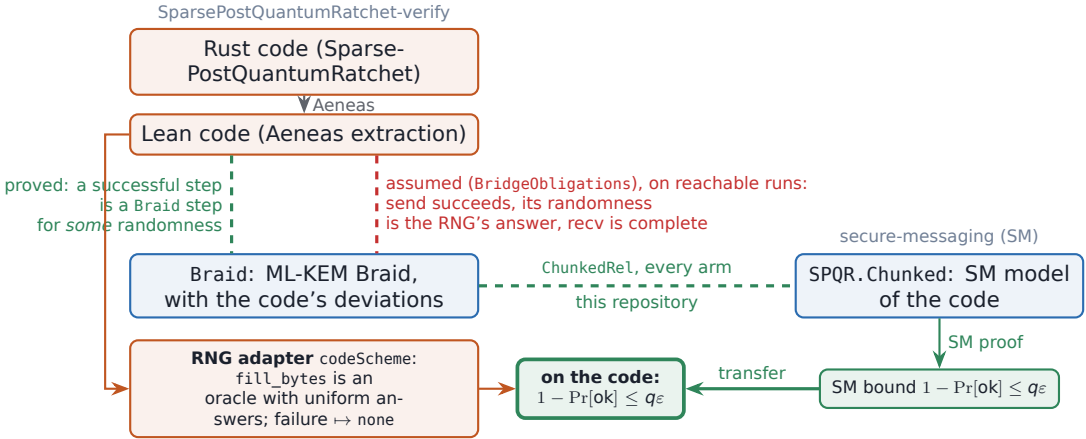
November 2026



Two proofs about SPQR, one gap



A conditional bridge across the gap



The bound needs the dashed arrow, not the proved one: "some randomness" says nothing about its distribution. Issues #785, #786 and #787 own the dashed arrow.

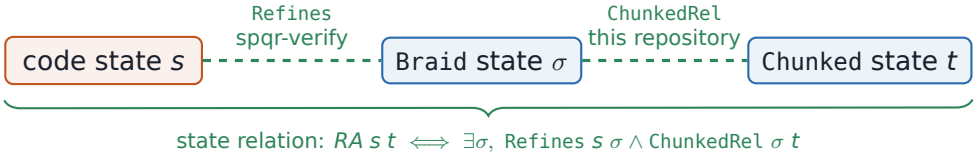
The result

Assume the four facts about the code in `BridgeObligations`, which are not proved yet. The send and receive facts cover reachable runs only. Let `adv` make at most q send queries, counted over both parties, with $q + 2 < 2^{64}$.
Then the probability that the correctness game fails on the code is at most $q\varepsilon$.

```
theorem code_correctness_error_le (hB : BridgeObligations)
    (sampleInitKey : ProbComp InitKey)
    (adv : SCKACorrectnessAdversary Message) (q : ℕ)
    (hq : SendQueryBound adv q) (hq64 : q + 2 < 2 ^ 64) :
    1 - Pr{let x ← correctnessExp (codeScheme sampleInitKey) adv}
          [x = true] ≤
      (q : ℝ≥0∞) * codeKem.correctnessError ProbCompRuntime.probComp
```

- `codeScheme`: the code as an SCKA scheme, with uniform bytes for each send. Under `IdealRng`, the bound holds for the extracted `send` (probability deck).
- ε: the correctness error of `codeKem`, the KEM of the code's ML-KEM models. It has no numerical bound yet.
- The adversary delivers unmodified recorded messages, in any order and with replay. No injected messages, no liveness.
- No `sorry`. Axioms beyond Lean's three: HKDF, HMAC, one libcrux workaround.

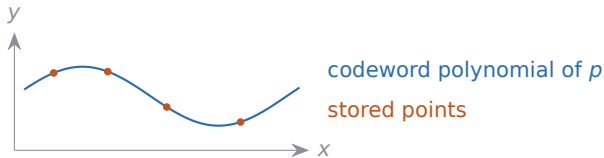
The chain of relations



- *RA*: a code state and an SM state are related through `Braid`. The code state refines some `Braid` state σ (spqr-verify), and σ is related to the SM state (`ChunkedRel`). Both parties use it.
- `ChunkedRel`: the same of the eleven states, equal epochs and authenticators, byte fields as typed values, equal decoders, and each code encoder's `u32` counter is SM's counter modulo 2^{32}. The header that `Refines` leaves free is taken from the SM state.
- *RM*: `Braid`'s message map `absMsg`, followed by the chunk map $c \mapsto (c.\texttt{index}, c.\texttt{data})$ into SM.
- `send_chunked` and `receive_chunked` cover every arm of both machines.

The Rust decoder meets SM’s decoder law

SM’s machines are generic in a `StreamDecoder`. SM’s own decoder decodes a set of chunks. The Rust decoder skips points as they arrive, so its state depends on the order. SM’s proofs need one law, `Lawful`: adding a chunk of p to a decoder that is honest for p keeps it honest, and it then decodes nothing or p.

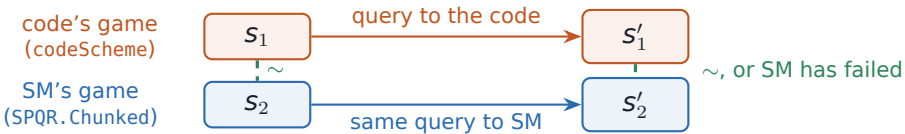


- The payload p gives 16 slot polynomials over $\mathrm{GF}(2^{16})$, each of degree $< K$.
- Honest state: every stored point lies on the slot polynomial of p. An honest chunk adds a point of the same polynomial, or is skipped.
- K points with distinct x fix a polynomial of degree $< K$, so a decoded value is p.
- `polyStreamDecoder_lawful`: one theorem for the four streams, $K \in \{3, 36, 30, 5\}$.
- The law is safety only: a decoder that never decodes meets it too.

Moving the bound to the code

SM's bound comes from `correctness_error_le_of_potential`, over an invariant Inv_2 and a potential V_2 on its game states. The bridge builds V_1 from V_2. Here $s_1 \sim s_2$ relates party states by *RA* and recorded messages by *RM*, with equal key tables, counters and flag.

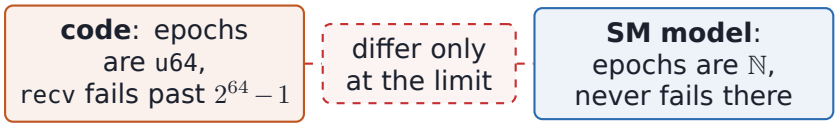
$$V_1(s_1) = \begin{cases} 1 & \text{if } s_1 \text{ has failed} \\ \min\big(1,\ \inf\{\, V_2(s_2) \mid s_1 \sim s_2,\ Inv_2\ s_2 \,\}\big) & \text{otherwise} \end{cases}$$



- One step, for *every* related s_2 with $Inv_2\ s_2$: $\mathbb{E}[V_1(s'_1)] \le \mathbb{E}[V_2(s'_2)] \le V_2(s_2) + c$, with $c = \varepsilon$ for a send and $c = 0$ otherwise. The first $\le$ holds because the states are related again, or SM failed and $V_2 \ge 1 \ge V_1$ (slide 9).
- The bound survives the infimum and the cap. No related s_2 gives $V_1 = 1$. No witness is chosen, and the games need not show the adversary equal values (`correctness_error_le_of_budgeted_transfer`).

Why 64-bit epochs never overflow

An adversary with fewer than $2^{64} - 2$ sends cannot push an epoch to the overflow point.



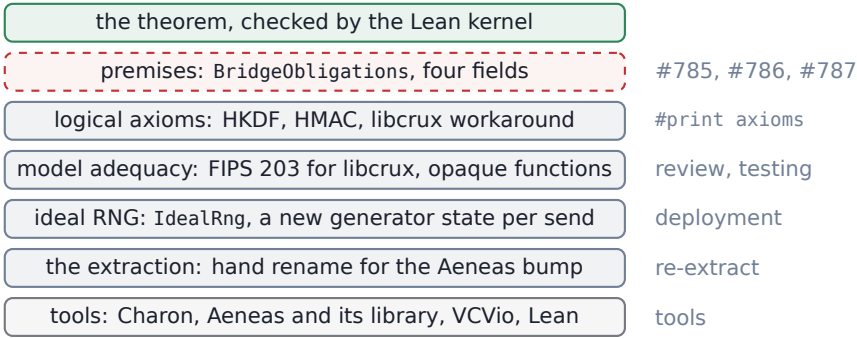
1. At the limit the two machines differ, so they cannot be related there.
2. Every epoch is at most the number of successful sends plus one (`codeGameInv_epochs`).
3. So the step obligations are needed only below a cap $B = 2^{64} - 2$ on the sends: fewer than B for a send, at most B for a receive, since deliveries go on after the last send. With $q + 2 < 2^{64}$, the adversary's q sends stay below B.

Derived transfer obligations

init	Every pair of initial states that the code can start in is related to a pair that SM can start in.
send	The two sends sample the same number of bytes: 64, 32 or none. They are coupled on equal samples, and their results are then related.
recv	If SM accepts a recorded message, the code accepts it too, with related results. If SM refuses, nothing is required.

- Key generation samples 64 bytes and encapsulation samples 32. For these two, `codeParams` builds SM's operations from the code's own primitives, so both sides use the same bytes.
- The other sends sample nothing. Both results are fixed and related.
- Why nothing is required when SM refuses: SM's game has failed already, and the bound counts that failure.

What the theorem trusts



- `#print axioms` audits only the axioms layer. The models, the opaque functions (SHA3, HMAC, libcrux sizes) and the RNG do not appear in it.
- A wrong model does not always break a premise: both sides of the bridge can agree on it. So the theorem is about the models, and their adequacy is a separate claim.
- ε is built from these models, so a bad primitive makes the bound weak, not false.

Status and next steps

piece	where	status
code refines `Braid`	spqr-verify	proved, one direction
generic stream decoders	SM	proved, merged
`Braid` related to `SPQR.Chunked`	this repository	proved, every arm
Rust decoder is lawful	this repository	proved
transfer of the bound	this repository	proved, for $q + 2 < 2^{64}$
extracted `send` with a generator	this repository	proved, under `IdealRng`
premise `mlkem_sizes`	spqr-verify, #786	open
premises `send_total`, `recv_complete`	spqr-verify, #785	open
premise `send_refines_bytes`	spqr-verify, #787	open
a number for ε	#788, #799, a model link	open

spqr-verify: SparsePostQuantumRatchet-verify. SM: secure-messaging.

Next

- Build the exact `BridgeObligations` record from upstream theorems (#785, #786, #787). Closing an issue is not enough if its statement differs.
- Bound ε: the #788 workaround, the `sampleNTT` cap (#799), and a proof that the two FIPS 203 models agree (README).

The probability deck

This deck gives the method and the result. One more deck, `slides/probability`, answers one question.

The extraction is deterministic, and SM bounds a probability. Where do the probabilities come from, and what must the code prove about its random bytes?